EVALUATION OF NLP SYSTEMS

Lecture 1 - Introduction

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Illustration by Freepik Stories

- what & why
- course organization
- know each other
- warm-up activity

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NLP system
or algorithm

many different purposes!

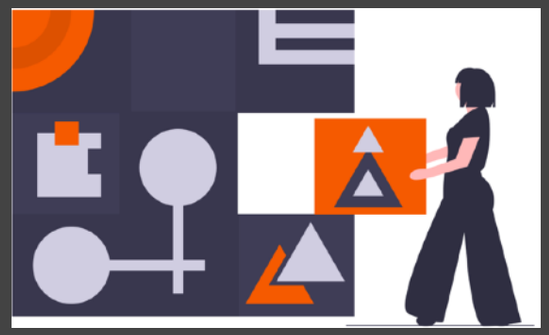
- machine translation
- dialogue
- speech recognition
- word segmentation
- semantic parsing
- document classification
- lemmatization
- stemming
- POS-tagging
- question answering
- language modeling
- named-entity recognition
- relation extraction
- coreference resolution
- automatic summarization
- semantic role labeling
- sentiment classification
- tokenization
- chatbot assistant
- taxonomy induction
- word sense disambiguation
- information retrieval
- text-to-speech
- syntactic parsing
- discourse analysis
- textual entailment
- grammar induction
- textual embeddings

- many more…

NLP system or algorithm

machine translation ⇢ language model
summarization ⇢ named-entity recognition
coreference resolution ⇢ word embeddings

can depend on other NLP systems



can be components of other systems

clinical diagnosis ⇢ information retrieval
robots ⇢ natural language understanding
customer assistance ⇢ sentiment analysis

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what does it do

what can go wrong

purpose

ethical issues

(NLP) system or algorithm

how to maintain it

computational aspects

costs

performance

who uses it

all these topics are relevant to evaluation, but our focus will be on measuring, analyzing and comparing performance

what does it do

what can go wrong

purpose

ethical issues

NLP system or algorithm

how to maintain it

substantial overlap with evaluation in machine learning and software development, but natural language has its specificities

computational aspects

costs

who uses it

how do we measure the quality of natural language outputs?

what happens if we change this or that?

are the users satisfied?

is the system doing what it should?

NLP system or algorithm

is system X better than system Y?

can the system be improved?

in which cases does it fail?

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Evaluation in NLP is trending…

EMNLP2020 Workshop

Evaluation and Comparison of NLP Systems (Eval4NLP)

Email: evaluation.nlp.workshop2020@gmail.com — Twitter: @NLPEvaluation

Overview

The submission deadline is extended to August 28rd. Dual submissions in the EMNLP main conference and our workshop are considered (Please withdraw the workshop paper by 15th, Sep if it is accepted at the main conference). Also notice that non-archival submissions are considered, but authors should inform us by email (there is no special button on the submission page).

Fair and adequate evaluations and comparisons are of fundamental importance to the NLP community to properly track progress, especially within the current deep learning revolution, with new state-of-the-art results reported in ever shorter intervals. This concerns the designing of adequate metrics for evaluating performance in high-level text generation tasks such as question and dialogue generation, summarization, machine translation, image captioning, poetry generation, etc.; properly evaluating word and sentence embeddings; and rigorously determining whether and under which conditions one system is better than another; etc.

dedicated workshop on EMNLP 2020

Beyond Accuracy: Behavioral Testing of NLP Models with CheckList

Marco Tulio Ribeiro	Tongshuang Wu	Carlos Guestrin	Sameer Singh
Microsoft Research	Univ. of Washington	Univ. of Washington	Univ. of California, Irvine
marcotcr@microsoft.com	wtshuang@cs.uw.edu	guestrin@cs.uw.edu	sameer@uci.edu

best paper award on ACL 2020

SemEval-2021

The 15th International Workshop on Semantic Evaluation

We are pleased to announce the following tasks for SemEval-2021!

TASKS

Websites and contact information for individual below - we will keep updating these as we receive them.

Link to CodaLab instructions, for both organizers and participants.

LREC Conferences

The International Conference on Language Resources and Evaluation is organised by ELRA biennially with the support of institutions and organisations involved in HLT.

theme of long-standing conference and workshop

Anya Belz @anyabelz · Aug 24

The #NLProc world will be a better place, really, if we all by default and at a minimum report experimental design, number of evaluators and items evaluated and how selected, significant differences, effect size and p-values. 2/2

Ehud Reiter @EhudReiter · Oct 11

Me: Your paper says A and B have 0.98 correlation. Seems surprising, why do you think this is the case?

Author: Whoops, actual correlation is 0.098, I left out a zero when copying the number into my paper.

I wonder how often this sort of thing happens...

Emily M. Bender
@emilymbender

Replying to @emilymbender

Furthermore, we can't let the current disaster scenario be used as a pass for valid evaluation of #NLProc systems. It's easy to throw ML methods at data; it's much harder to design the careful studies required to see if the ML methods actually produced useful output. >>

9:08 PM · Mar 24, 2020 · TweetDeck

Verena Rieser
@verena_rieser

Issues with papers on Machine Learning in NLP:
(1) Poor evaluation
(2) Inappropriate training data
(3) Incorrect presentation of related work
(4) Incremental research

Ehud Reiter @EhudReiter · Jun 6, 2018
New blog: Many Papers on Machine Learning in NLP are Scientifically Dubious
ehudreiter.com/2018/06/06/tru...

Nasrin Mostafazadeh @nasrinmmm · Jul 8

Amazingly both #acl2020nlp best paper & runner-up are on better **evaluating** #NLProc models. As less-rewarding & exhausting as it can sometimes feel to work on metricing, as opposed to developing better (&bigger) models,rethinking how we measure progress is more important than ever

Source: Twitter

Figure 1. *Salient events related to evaluation mentioned in this article (for evaluation campaign series, e.g. like TREC, only the first event is mentioned).*

Source: Paroubek et al. (2007)

...but has been important for decades.

What we'll talk about

- metrics
- data annotation
- best practices
- procedures
- statistical significance testing
- paradigms
- facets of evalution in different NLP tasks

- ☑ what & why
- ☐ course organization
- ☐ know each other

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WHAT THIS COURSE IS *NOT*

a one-way lecture where a professor pours knowledge into your heads and you output everything back in a final exam

WHAT THIS COURSE IS

minds on, interactive learning

a brainstorming-oriented seminar to train critical thinking in a safe environment

technical knowledge easily available nowadays with online slides, tutorials, videos etc.

let's focus on sharing and discussing ideas with each other: you will build the learning opportunities, I will help organizing and orienting this

Illustrations by unDraw

Part I

before Christmas break

- pre-recorded overview uploaded every weekend
- online meetings on Wednesdays at 12:15 (QA session and small groups discussion)
- worksheet answers must be sent before following meeting

Part II

after Christmas break

- your presentations: evaluation in different NLP tasks
- will publish details in ~2 weeks, once I have an idea of how many students will take the course

Part III

semester break

- final project
- three options: traditional, practical or creative
- preliminary guidelines on website, but there will be a meeting to discuss the details

Illustration by Freepik Stories

Grading

- 20%: active participation in the course (contributing to online group discussions and handing in exercises)
- 40%: presentation
- 40%: final project

To pass the course, you must achieve a performance of at least 70% **on each of the items above separately**.

Exercises: I will not grade them, because in general there will be no right and wrong. But for a submission to be considered valid, your answers must be coherent and complete (no exercise left blank); that I will check.

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- check the guidelines very carefully
- stay tuned on Moodle for important announcements
- use the Moodle forum
- feel free to give me feedback and suggestions
- if you have any problem (health, family etc) that will affect your participation, talk to me asap

ANY QUESTIONS?

Illustration by undraw

- ☑ what & why
- ☑ course organization
- ☐ know each other

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Who are you?

- name
- course & semester
- interests
- why you are here
- experience with this topic
- expectations

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Virtual Evaluation Hunt

1 3 women who publish about evaluation in NLP	2 an image about the METEOR metric	3 who invented the BLEU score?	4 a company advertising evaluation	5 someone in Twitter complaining about data annotation
6 5 evaluation metrics	7 an image about k-fold validation	8 who proposed the kappa metric for inter-annotator aggrement?	9 a GitHub repository with an evaluation tool	10 someone in Twitter unhappy about an NLP metric
11 2 events about evaluation	12 an image about hypothesis testing in NLP	13 who was the head of the EAGLES project?	14 slides of a university lecture about evaluation	15 someone in Twitter giving advice about evaluation
16 10 published papers about evaluation	17 an image about linguistic error analysis	18 who is organizing the Eval4NLP workshop in EMNLP?	19 a video tutorial about NLP evaluation	20 someone in Twitter promoting a good evaluation practice

See you next week!

- ☑ what & why
- ☑ course organization
- ☑ know each other
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